

Sponsoring Organisation:	Implementation Date:	01 December 2015
DH Informatics Directorate	Subject: Time Display for Clinical Systems within the NHS in England	
DATA SET CHANGE NOTICE		
This DSCN informs users of the approval of a new information standard by the Information Standards Board for Health and Social Care (ISB). It was presented at ISB’s meeting on 24 September 2009 and subsequently approved having met the required conditions.		
Summary: This user interface standard defines the valid display values for exact time, approximate time, time duration and display of date and time combinations. Organisations developing or upgrading applications for which the standard is in scope MAY implement the time display standard immediately and SHOULD do so at their next major system release. All clinical systems within the NHS in England MUST conform to the time display standard by 01 December 2015. Patient safety risks associated with the time display standard are specified in the safety case accompanying the standard. Organisations that develop, supply, deploy and use clinical systems must identify and manage any risks in current systems and mitigate implementation risks for which responsibility for mitigation has been transferred to the supplying or deploying organisation. The process for managing safety risks is detailed in NHS information standards for the application of patient safety risk management to the manufacture, deployment and use of health software. The standard can also be used to inform design of user interfaces in other systems and devices provided any safety risks are identified and managed, for example, risks related to reduced screen size.		
Other Datasets / Return Affected: None		
Related DSCNs: ▪ DSCN 18/2009 - Application of Patient Safety Risk Management to the Deployment and Use of Health Software. ▪ DSCN 14/2009 - Application of Patient Safety Risk Management to the Manufacture of Health Software. ▪ AN 22/2007 – Date and Time Display and Input for Clinical Systems in the NHS in England. ▪ DSCN 09/2010 – Patient Banner for Clinical Systems within the NHS in England. ▪ DSCN 07/2010 – Date Display for Clinical Systems within the NHS in England. ▪ DSCN 02/2010 – Date and Time Input for Clinical Systems within the NHS in England.		
Impact of Change:		
Service:	Major	System Suppliers: Major
The Information Standards Board for Health and Social Care (ISB) is responsible for approving information standards.		

DATA SET CHANGE NOTICE

Reference No:	DSCN 08/2010
Version No:	1.0
Subject:	Time Display for Clinical Systems within the NHS in England
Type of Change:	Introduction of a new approved Information Standard
Implementation Date:	01 December 2015
Business Justification:	To make display and user interaction consistent across all clinical systems within the NHS in England in order to increase patient safety by maximising clinical utility and minimising reading and transcribing errors.

Introduction

Displaying and inputting unambiguous information in a consistent format is a core element in ensuring effective patient care. It is vital that healthcare IT users correctly interpret and input information relating to all aspects of patient care and management.

This user interface standard defines the valid display values for exact time, approximate time, time duration and display of date and time combinations.

The standard has been developed as part of the Common User Interface (CUI) programme within the Technology Office of the Department of Health Informatics Directorate. This programme is addressing the safety risks posed by variation across the NHS in the way information is input and displayed. Its aim is to support healthcare IT users in correctly inputting and interpreting information relating to all aspects of patient care and management.

Background

The display of exact times, approximate times and durations within software applications has inherent risks of misinterpretation based on how these values are displayed. Currently, there is no specific time display standard within the NHS for clinical applications, implying that various systems across the NHS will implement different approaches to displaying time. Because of this inconsistency, risks to effective clinical care and to patient safety arise.

Currently, clinical and healthcare IT systems used across all care settings in the NHS in England differ in the way that time is displayed. For example, our review of clinical systems showed a mix of both 24-hour and 12-hour time display formats being used. Inherent within this is the risk that healthcare professionals moving between clinical systems made by different suppliers can misinterpret time, leading to patient safety incidents, as defined by the National Patient Safety Agency (NPSA). This risk is particularly relevant to healthcare practitioners who trained outside of the UK, since they may well be used to different conventions.

The safety case and guidance accompanying the standard provide evidence of clinical safety issues and there is a clear argument that unambiguous time display would reduce the potential for human error leading to patient safety incidents.

Details of the Standard

The standard is published with associated guidance in a Common User Interface Design Guide. It is applicable to user interfaces displayed on desktop or laptop computers. It is assumed that, at a minimum, these computers are capable of operating at a minimum display resolution of 1024 x 768, and have a keyboard and pointing device.

The following items are covered in the guidance:

- Exact time – the precise time of an event.
- Approximate time (or “fuzzy” time).
- Time duration – a period of time.
- Display of date and time combinations.

Aspects that are **out of scope**:

- Time entry.
- Durations shorter than a second.
- Dates (display and input).
- Labels.
- Synchronisation.
- Data storage
- Display styles.

Safety Case

Clinical safety requirements and objectives related to the display of times in clinical systems are documented in the safety case that accompanies this standard (www.isb.nhs.uk/documents/cui). Organisations that develop, supply, deploy and use clinical systems are required to identify and manage any risks in current systems until the standard can be implemented. They must also mitigate those live implementation risks outlined in the safety case for which responsibility for mitigation has been transferred to the supplying or deploying organisation.

The process for assessing and managing safety risks is detailed in NHS information standards for the application of patient safety risk management to the manufacture, deployment and use of health software (<http://www.connectingforhealth.nhs.uk/dscn/dscn2009/data-set-change/>), i.e.:

- Application of Patient Safety Risk Management to the Manufacture of Health Software (DSCN 14/2009); and
- Application of Patient Safety Risk Management to the Deployment and Use of Health Software (DSCN 18/2009).

Conformance

For full conformance to be demonstrated, the elements marked as ‘Mandatory’ in the Common User Interface Design Guide must be complied with and there must have been mitigation of those live implementation risks outlined in the safety case for which responsibility for mitigation has been transferred to the supplying or deploying organisation.

Timescales for Implementation

FRAMEWORK	Health and Social Care Personnel	Organisation	IT Suppliers
Effective Date “may use”	Immediate	Immediate	Immediate
Implementation Date “must use”	When available in system	At next major system release	At next major system release
Conformance Date “must be used effectively and assessed for use”	01 December 2015	01 December 2015	01 December 2015
Superseded Date (of prior standard) “stop using prior standard”	Not applicable	Not applicable	Not applicable

Effects on Other Information Standards:

None

Sponsor Details

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Technology Office

Further Information and Support

- CUI Time Display Design Guide - www.cui.nhs.uk
- Clinical Safety Case - www.isb.nhs.uk/documents/cui
- CUI website - www.cui.nhs.uk
- Standards documentation – www.isb.nhs.uk/documents/cui
- Queries - cuistakeholder.mailbox@nhs.net

Appendix

This DSCN must be read in conjunction with the Clinical Safety Case and Closure Report for Demographics Information Input and Display, which can be found at the following link:
www.isb.nhs.uk/documents/cui

Table Notes:

1. Relevant organisations are those organisations as defined in the standard who must take direct action to implement the standard
2. IT Suppliers are all suppliers to the organisations listed at ¹ who supply functionality pertinent to that standard
3. **Effective Date** is the date from which a new standard can be used but may not be mandatory. This might facilitate piloting, for example, or enable time for system functionality development. At this point, **you “may use” the standard.**
4. **Implementation Date** is the point from which the new standard becomes mandatory. Ideally, it inherently implies organisations use appropriate systems i.e. the date is the same for organisations and suppliers. However, there may be circumstances where interim workarounds are required i.e. the date is different for organisations and suppliers. At this date, **you “must use” the standard.** Where the standard demands data is submitted centrally, sub components of implementation date (and possibly 'effective date') are:
 5. **Collection Start Date** – this is the date collection of data must begin
 6. **First Submission Date** – this is the date of first submission of data centrally
 7. **Reporting Period / Submission Cycle** – If the standard calls for further collection and submission at defined intervals, this cell provides text of the reporting period (e.g. calendar month, financial year) and the submission cycle (e.g. submit data monthly on the 10th working day of the subsequent month).
8. **Conformance Date** is the date from which the service and IT system suppliers must use the standard as envisaged i.e. using appropriate IT solutions rather than interim workarounds and, if the standard requires it, an independent, authoritative body or legitimate internal audit would conduct a conformity assessment with the expectation of full conformance by all relevant parties. It is the **“must use standard effectively and assessed for use”** date
9. **Superseded Date** of the prior standard sets the date at which the prior standard is replaced by the new standard i.e. the prior standard must no longer be used. This date will apply only where there was a pre-existing standard made redundant by the new standard. It might be different from preceding dates in the framework if, for example, a new and old standard run in parallel for a period. It is the date from which you **“stop using the prior standard”.**